ADITYA KANTH MANNE

AI/ML ENGINEER | QUANTITATIVE RESEARCH | FINANCIAL MARKETS

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SUMMARY

Ph.D. student in Artificial Intelligence with an M.S. in Engineering Management and Data Analyst experience, applying Python and statistical methods to financial-market research and AI evaluation. Built a BTCUSDT spot–perpetual workflow using Binance spot and futures data for 1–10-second lead–lag research, with timestamp controls and execution-cost simulation; developed an offline RAG QA demo with retrieval baselines and citation/abstention checks. Seeking Summer 2027 quantitative research/analytics or AI/ML internships in banking and investment firms.

TECHNICAL SKILLS

Quant Research	Probability and statistics, hypothesis testing, Bayesian inference, linear algebra, optimization, stochastic calculus, financial econometrics, asset pricing, factor models, derivatives pricing, portfolio optimization, risk modeling, time-series modeling, Monte Carlo simulation, market microstructure, Heston option pricing, implied volatility, alpha research, backtesting, transaction costs, market impact, MATLAB.
AI & Machine Learning	Supervised learning, regression, classification, clustering, gradient boosting, XGBoost, feature engineering, model evaluation, PyTorch, deep learning, transformer models, embeddings, LLM and RAG systems, BM25, TF-IDF/SVD, citation and abstention evaluation, model serving, MLOps.
Engineering & Data	Python, R, SQL, C++, scikit-learn, pandas, NumPy, Polars, PyArrow, FastAPI, Airflow, Spark, MLflow, LangGraph, Qdrant, REST APIs, ETL, YAML, schema monitoring, SQL Server, PostgreSQL, Git, Linux, pytest, AWS.
Analytics	Tableau, Power BI, Amazon QuickSight.

EXPERIENCE

Early Learning Coalition of Flagler & Volusia | Data Analyst — **JUL 2024 – PRESENT**

- With state-source records needing a nightly CRM refresh, built an incremental Python ETL through SQL Server to WebAuthor; delivered a repeatable source-to-CRM flow.
- As vendor APIs exposed changing fields, added YAML-driven ETL settings and schema-drift monitoring; made integrations configurable and surfaced source changes for review.
- To align finance and program reporting, reconciled General Ledger, provider, and enrollment datasets across **3 data domains**; supplied a consistent view for downstream analysis.
- For monthly provider oversight, built Tableau Provider Analytics views for monitoring stages and outcomes, progressive enforcement, and Gold Seal counts by year; consolidated recurring indicators.
- To explain program and compliance patterns, reported SR/VPK reasons across **3 fiscal years** and the **top 5 DCF violations**; converted the Excel macro workflow to Python for repeatable analysis.

PROJECTS

BTC Perpetual–Spot Lead–Lag & Maker Quoting | *Python, pandas, NumPy, scikit-learn* — **2026**

- To test market price discovery, built a BTCUSDT data workflow for top-of-book quotes and trades from **2 Binance venues** (spot and futures); evaluates lead–lag over **1–10-second horizons**.
- Because venue clocks can diverge, standardized receive-time alignment and added clock-gap diagnostics, a delayed-perpetual placebo, and mirrored-venue checks; tests whether apparent lead survives timestamp controls.
- To validate pipeline behavior before real-market analysis, implemented synthetic lead, null, and delayed-feed cases; **all 30 tests pass** on synthetic data.
- To measure incremental prediction value, compare spot-only Ridge with spot-plus-perpetual Ridge/GBDT using a **60/20/20 chronological whole-day split**; report day-level out-of-sample R^2 uplift over the 1–10-second horizons.
- To evaluate trading value after execution frictions, simulate mid, spot-only, and full fair-value quotes across **5 latency settings** (5, 20, 50, 100, 250 ms) and 1/5/30-second markouts; estimate breakeven maker fees and inventory exposure.

RAGFix | Retrieval Failure Diagnosis | *Python, BM25, TF-IDF/SVD, RAG evaluation* — **2026**

- With technical answers varying by document version, implemented an offline end-to-end QA demo that ingests versioned Markdown and chunks by headings; produces traceable retrieval and answer records for failure review.
- To compare retrieval strategies, implemented BM25 and TF-IDF/SVD offline baselines and exposed dense/hybrid retrieval options with version filters and reciprocal-rank fusion; makes retrieval approaches configurable for evaluation.
- To prevent unsupported answers, added citation-ID and abstention validators plus a post-check that converts uncited responses to an explicit abstention; supplies deterministic grounding guardrails.
- To localize failures before selecting a fix, built diagnosis and reporting across corpus, retrieval, prompt, and model stages; directs reviewers toward the stage that needs intervention.
- To compare fixes without tuning on the holdout, designed development, regression, and locked splits with paired bootstrap confidence intervals, McNemar tests, failure rates, token cost, and p50/p95 latency; protocol targets approximately **600 baseline answers**, **200 failure labels**, **5–8 categories**, and a separate **100-question** fresh-data set.

EDUCATION & CERTIFICATIONS

University of the Cumberlands | Ph.D. Student, Artificial Intelligence — **2026 – PRESENT**

University of South Florida | M.S. Engineering Management | GPA: 3.83/4.00 — **2022 – 2023**

AWS Certified Machine Learning Engineer – Associate | Valid through May 2028